Construction Chemicals

SUBMITTAL DOCUMENT

Leakinject PU

Solvent-Free PU Injection Resin
for Stopping Active Leaks



Active Leaks
Stops water

Expansion
Up to 30×

Foam
Closed-cell

Resistant
Acids & fuels

Project	
Main Contractor	
Consultant	
Submittal No. / Rev.	00
Date	Nov 2026

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METHOD STATEMENT

Leakinject PU — Solvent-Free PU Injection Resin for Stopping Active Leaks

Project Name		**Document No.**	OMX-MS-LEAKINJECTPU
Main Contractor		**Revision**	01
Consultant		**Date of Issue**	Nov 2026
Product	Leakinject PU	**Status**	For Approval

1. Purpose

This method statement defines the procedure, sequence, resources, quality checks and safety measures for the supply and application of Leakinject PU on site. It ensures that the work is carried out safely, in line with the approved drawings and project specification, and achieves the performance stated in the O MIX Technical Data Sheet.

2. Scope of Work

This method statement covers preparation, mixing, application, curing and inspection of Leakinject PU (solvent-free pu injection resin for stopping active leaks).

- Inspection and acceptance of the substrate and the work area.
- Surface preparation and repair of defects.
- Priming of the prepared surface.
- Mixing and application of the material as described in this document.
- Curing, protection and handover of the completed area.
- Quality control, inspection and documentation of each stage.

3. Definitions

Term	Meaning
Pot life	Time the mixed material remains workable after Part A and Part B are combined.
Recoat interval	Minimum and maximum time between coats to achieve inter-coat adhesion.
WFT / DFT	Wet film thickness measured during application / dry film thickness after cure.
Dew point	Temperature at which moisture condenses; the substrate must be at least 3 °C above it.
Kit / set	Pre-weighed Part A and Part B supplied to be mixed completely.

4. Roles & Responsibilities

Role	Responsibility
Project Manager	Overall responsibility for planning, resources, approvals and coordination with the consultant and main contractor.

Role	Responsibility
Site Engineer	Supervises the work daily; confirms substrate readiness, climate conditions, mixing, consumption and sequence; keeps daily records.
QA/QC Inspector	Carries out inspections and tests at each stage, records results, raises inspection requests at hold points and signs the checklist.
HSE Officer	Conducts toolbox talks, checks PPE, ventilation, access and safe storage; stops work if unsafe conditions arise.
Foreman / Applicators	Carry out the work as per this method statement and report any deviation immediately.
Storekeeper	Receives, stores and issues materials in original packaging under the stated storage conditions; controls batch numbers.

5. Materials

Material	Description
Leakinject PU	Resin 20 kg jerrycans (also IBC 1050 kg); catalyst 2 kg and 20 kg.

5.1 Key product data

Property	Typical Value
Resin	Brown liquid, approx. 120 mPa·s
Density	1.18
Flash point	> 148 °C
Free expansion	up to 30 times
Compressive strength (foam)	11.5 MPa
E-modulus	295 MPa
Catalyst dosage	As recommended for site conditions

6. Tools & Equipment

- Grinder / shot blaster or blasting unit
- Industrial vacuum
- Slow-speed mixer
- Rollers
- Brushes
- Airless spray where specified
- Wet-film comb
- DFT gauge
- Thermo-hygrometer and surface thermometer (dew point check)
- Moisture meter
- Wet-film thickness gauge
- Measuring buckets and scale

- Masking tape and protective sheeting
- Lighting and extension cables

6.1 Personal protective equipment

- Chemical-resistant gloves
- Goggles
- Overalls
- Respirator where required

7. Storage & Handling

- 12 months in original, unopened packaging. Store dry at +5 °C to +35 °C, away from direct sunlight.
- Store on pallets, off the ground, in a dry, covered and ventilated area.
- Keep containers closed until use; use materials on a first-in, first-out basis.
- Keep Part A and Part B of each kit together and do not mix parts from different batches.
- Store away from heat, sparks and ignition sources.
- Handle with care to avoid damage to packaging; do not stack beyond the safe height.

8. Pre-Start Requirements

- Material submittal, this method statement and the inspection plan approved by the consultant.
- Work area handed over clean, free of other trades and with safe access.
- Adequate lighting, power supply and ventilation available.
- Climate conditions checked and recorded before work starts.
- A trial area prepared and approved where required by the project.
- Toolbox talk held with the work team covering this method statement and the safety measures.

9. Surface Preparation

1. Locate leaks and mark crack lines.
2. Drill injection holes and install packers.
3. Flush holes with water.
4. Measure substrate temperature, air temperature and relative humidity; confirm the substrate is at least 3 °C above dew point.
5. Protect adjacent surfaces, drains and fixtures with masking tape and sheeting.
6. Obtain QA/QC approval of the prepared surface before starting the next step.

10. Mixing

1. Condition materials to between +15 °C and +25 °C before mixing.
2. Check the batch numbers and expiry dates of both parts and record them.

3. Add catalyst to the resin as recommended.
4. Mix thoroughly; use within the working time.
5. Mix only the quantity that can be applied within the pot life.

11. Application

1. Drill holes at 45° to intersect the crack at 15 – 30 cm spacing, staggered on both sides.
2. Install packers and flush holes with water.
3. Inject resin from the lowest packer at low pressure, increasing as needed.
4. Move to the next packer when resin appears.
5. Work in manageable bays, keeping a wet edge to avoid lap marks.
6. Check consumption by area and wet film thickness regularly during application and record the results.

12. Curing & Protection

- Allow foam to cure; remove packers and seal holes with repair mortar.
- Protect the applied area from water, dust, other trades and traffic until the stated cure times are reached.
- Barricade the area and display warning signs.

13. Quality Control & Inspection

Check	Stage	Acceptance Criteria	By
Substrate	Before application	Prepared, sound and clean as specified	QA/QC
Climate	During work	Material and substrate temperature: above +5 °C	Site Engineer
Consumption	During application	0.2 – 2 kg per linear metre of crack	QA/QC
Finish	After curing	Uniform, no defects	QA/QC

13.1 Inspection and test plan

Activity	Inspection / Test	Frequency	Record
Material receipt	Product, batch, expiry, packaging condition	Each delivery	Material inspection report
Climate	Air / substrate temperature, RH, dew point	Start of shift and every 4 hours	Daily climate log
Surface preparation	Visual, moisture, soundness	Each area	Inspection request (hold point)
Application	Consumption, thickness, appearance	Each coat / area	Daily progress report
Final inspection	Visual, continuity, finish	Each completed area	Handover checklist

14. Defects & Remedies

Defect	Likely Cause	Remedy
Pinholes / bubbles	Outgassing from porous substrate; air entrapped in mixing	Seal with primer; mix at low speed; de-aerate with spiked roller; sand and recoat
Blushing / tacky surface	High humidity or low temperature during cure	Wash and abrade; recoat in suitable conditions
Poor adhesion	Contaminated, weak or damp substrate; recoat window exceeded	Remove, re-prepare surface and reapply
Soft spots	Incorrect mixing or ratio	Remove affected area and reapply with full kits
Uneven colour / lap marks	Loss of wet edge; different batches	Abrade and apply a further coat from one batch

15. Health, Safety & Environment

Hazard	Control Measures
Skin and eye contact	Gloves, goggles, overalls; eye-wash available
Inhalation of vapours / dust	Ventilation; masks as required
Slips and trips	Housekeeping; barricades and signs
Manual handling	Correct lifting technique; mechanical aids
Electrical tools	Inspected tools; protected cables

- Wear chemical-resistant gloves, safety goggles and overalls.
- Uncured resin may cause skin sensitisation; avoid skin and eye contact.
- Ensure good ventilation in enclosed areas.
- Eyes: rinse immediately with plenty of clean water for at least 15 minutes and seek medical advice.
- Skin: remove contaminated clothing and wash with soap and water; do not use solvents on skin.

15.1 Waste & environment

- Collect empty containers and residues in designated bins.
- Do not discharge material or washings into drains or soil.
- Allow small quantities of leftover mixed material to cure before disposal; dispose of according to local regulations.

16. Cleaning

- Clean pumps and tools with PU cleaner immediately after use.

17. Limitations

- Material and substrate temperature: above +5 °C.
- Water can be present (reaction needs water).
- Do not apply in rain or when rain is expected before the material has set.
- Any deviation from this method statement must be approved by the Site Engineer and the consultant.

Appendix A — Inspection Checklist

Inspection Item	Result	Remarks
Substrate prepared and approved	☐ Yes ☐ No	
Batch numbers recorded	☐ Yes ☐ No	
Material mixed as specified	☐ Yes ☐ No	
Applied at specified consumption	☐ Yes ☐ No	
Curing and protection carried out	☐ Yes ☐ No	
Final inspection passed	☐ Yes ☐ No	

Inspected by	Signature	Date

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Technical Data Sheet

Leakinject PU — Submittal Document

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Product Datasheet
Edition: 01 | Date: Nov 2026
Waterproofing

Leakinject PU

Solvent-Free PU Injection Resin
for Stopping Active Leaks



Product Description

Leakinject PU is a solvent-free polyurethane injection resin that reacts with water to form a semi-rigid, closed-cell foam. It stops active water leaks under high pressure, expands up to 30 times when free, does not shrink, resists acids and fuels, is non-toxic and suitable for potable-water structures.

Uses

- Stopping active leaks in cracks and joints.
- Basements, tunnels and shafts.
- Water tanks and manholes.
- Sealing around pipe penetrations.
- Potable water tanks and structures.

Advantages

- Reacts with water; stops active leaks.
- Free expansion up to 30 times.
- Closed-cell, non-shrinking foam.
- Resistant to acids and fuels.
- Solvent-free and non-toxic.

Application

1. **Preparation**
   Drill injection holes along the crack and install packers.
2. **Mixing**
   Add catalyst (CAT) to the resin as recommended for site conditions.
3. **Injection**
   Inject with a single-component pump from the lowest point upward.

Consumption

0.2 – 2 kg
per linear metre of crack
Depends on crack width and voids.

Packaging

Resin 20 kg jerrycans (also IBC 1050 kg); catalyst 2 kg and 20 kg.

Storage & Shelf Life

12 months in original, unopened packaging. Store dry at +5 °C to +35 °C, away from direct sunlight.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
Resin	Brown liquid, approx. 120 mPa·s	—
Density	1.18	—
Flash point	> 148 °C	—
Free expansion	up to 30 times	—
Compressive strength (foam)	11.5 MPa	—
E-modulus	295 MPa	—
Catalyst dosage	As recommended for site conditions	—

Consumption

Crack condition	Consumption
Fine cracks	0.2 – 0.5 kg/m
Wide cracks / voids	0.5 – 2 kg/m

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- Material and substrate temperature: above +5 °C.
- Water can be present (reaction needs water).

Application Notes

- Start at the lowest point and move upward.
- Inject until resin appears at the next packer.
- Adjust catalyst dosage for reaction speed.
- Clean the pump with PU cleaner after use.

Health & Safety

- Wear chemical-resistant gloves, safety goggles and overalls.
- Uncured resin may cause skin sensitisation; avoid skin and eye contact.
- Ensure good ventilation in enclosed areas.

First Aid & Cleaning

- Eyes: rinse immediately with plenty of clean water for at least 15 minutes and seek medical advice.
- Skin: remove contaminated clothing and wash with soap and water; do not use solvents on skin.
- Inhalation: move to fresh air; seek medical advice if symptoms persist.
- Ingestion: do not induce vomiting; seek medical advice immediately.
- Clean pumps and tools with PU cleaner immediately after use.

The information in this datasheet is based on laboratory tests and practical experience and is given in good faith. Users should verify product suitability for their specific application. O MIX reserves the right to update this information without prior notice.

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Test Reports

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To be attached: laboratory test reports supporting the values in the Technical Data Sheet.

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Approvals & Certificates

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To be attached: approvals and certificates, where available.